

DSA 8020 Project 1 Makeup

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```
library(ggplot2)
library(MASS)
library(mgcv)
```

```
## Loading required package: nlme
```

```
## This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.
```

```
library(car)
```

```
## Loading required package: carData
```

Problem 1: Gala

```
library(faraway)
```

```
##
```

```
## Attaching package: 'faraway'
```

```
## The following objects are masked from 'package:car':
```

```
##
```

```
## logit, vif
```

```
data(gala)
```

```
# Excluding Species from analysis
```

```
gala <- subset(gala, select = -Species)
```

```
head(gala)
```

```
##           Endemics  Area Elevation Nearest Scruz Adjacent
## Baltra           23 25.09      346      0.6   0.6    1.84
## Bartolome        21  1.24      109      0.6  26.3   572.33
## Caldwell          3  0.21      114      2.8  58.7    0.78
## Champion          9  0.10       46      1.9  47.4    0.18
## Coamano           1  0.05       77      1.9   1.9   903.82
## Daphne.Major     11  0.34      119      8.0   8.0    1.84
```

2. Is at least one of the predictors useful in predicting the response (Endemics)?

Establishing an alpha (α) value of 0.05.

Code:

```
# getting p-val
```

```
model <- lm(Endemics ~ Area + Elevation + Nearest + Scruz + Adjacent, data = gala)
```

```
summary(model)
```

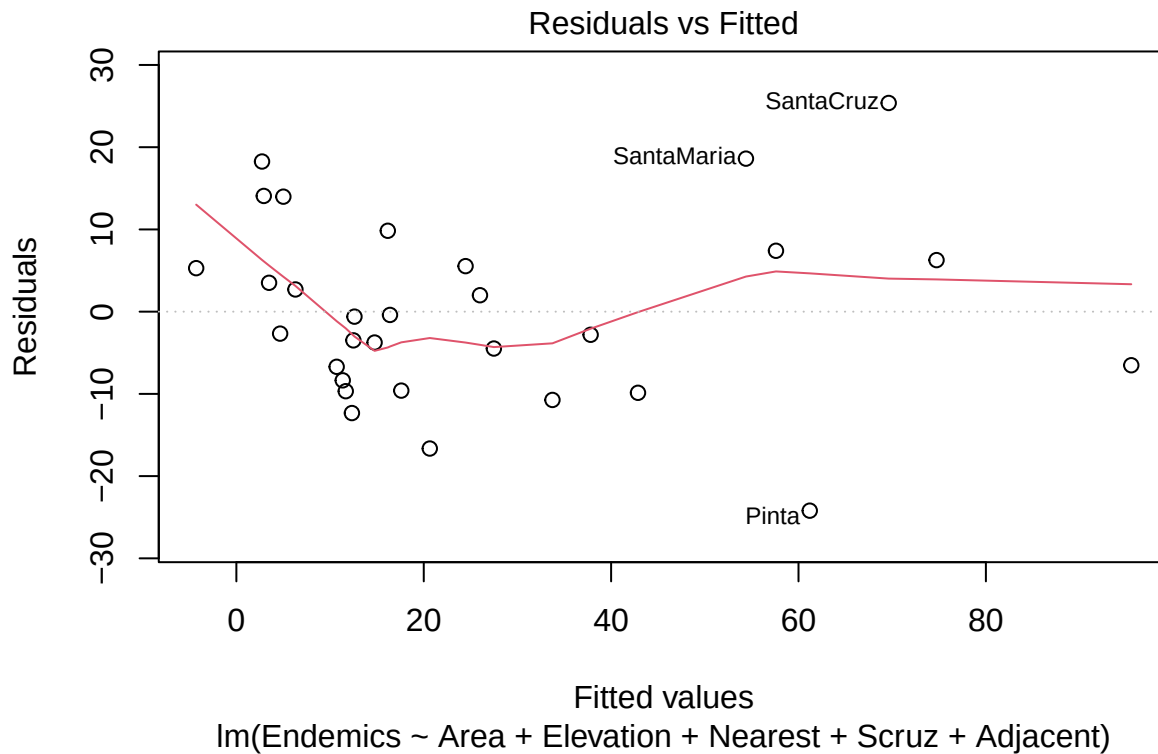
```
##
## Call:
## lm(formula = Endemics ~ Area + Elevation + Nearest + Scruz +
##      Adjacent, data = gala)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -24.201  -7.941  -1.637   6.086  25.376
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.099665   3.859507   1.321   0.1989
## Area        -0.008466   0.004518  -1.874   0.0732 .
## Elevation     0.083530   0.010813   7.725 5.83e-08 ***
## Nearest       0.025173   0.212405   0.119   0.9066
## Scruz        -0.056623   0.043403  -1.305   0.2044
## Adjacent     -0.017438   0.003567  -4.889 5.50e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.29 on 24 degrees of freedom
## Multiple R-squared:  0.8328, Adjusted R-squared:  0.7979
## F-statistic: 23.9 on 5 and 24 DF, p-value: 1.35e-08
```

```
# correlation
```

```
cor(gala[, c("Area", "Elevation", "Nearest", "Scruz", "Adjacent")], gala["Endemics"])
```

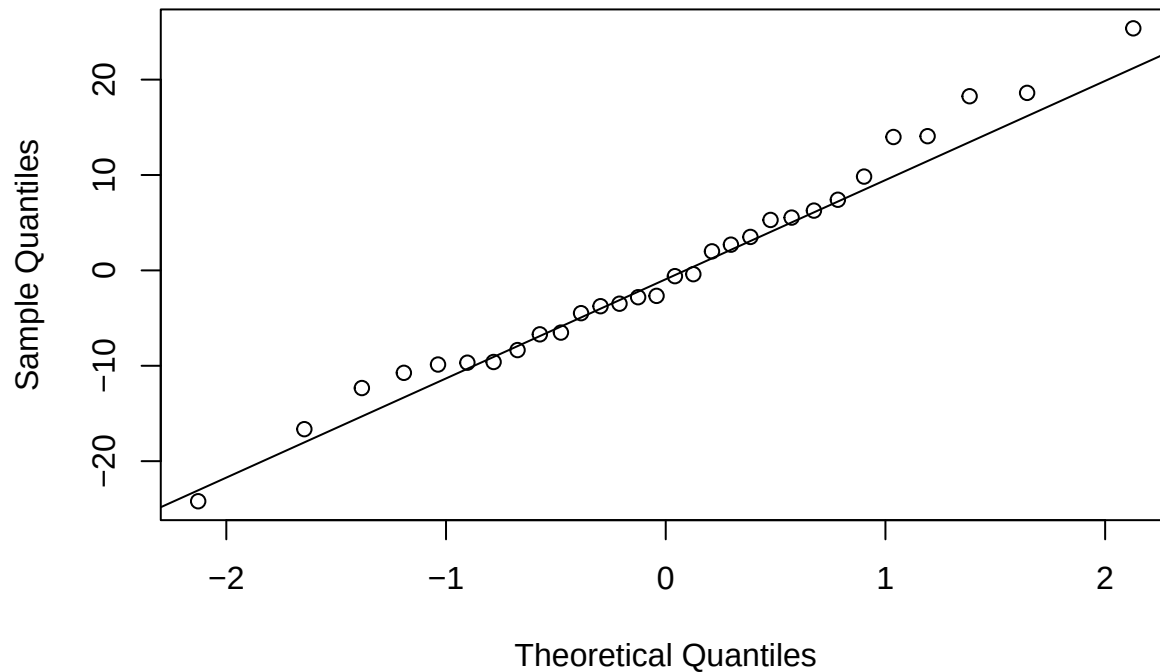
```
##           Endemics
## Area      0.616979087
## Elevation 0.792904369
## Nearest   0.005994286
## Scrutz    -0.154264319
## Adjacent  0.082658026
```

```
# Checking model assumptions
# residual vs fitted
plot(model, which = 1)
```



```
# qq plot
qqnorm(resid(model))
qqline(residuals(model))
```

Normal Q-Q Plot



Answer:

- Allowing an $\alpha = 0.05$, we can see that...
 - Elevation (p-val: 5.83e-08) and Adjacent (p-val: 5.50e-05) are useful in predicting the response (Endemics). Elevation (0.79) has high Pearson correlation with Endemics signifying a strong linear relationship. However, Adjacent (0.08) has a relatively low Pearson correlation with Endemics, while it still may have a significant relationship with the response, it is unlikely that this relationship is linear. Alternatively Adjacent can be acting as a supressor variable, improving prediction accuracy when combined with other predictors like Elevation.
 - Area (p-val: 0.0732) is slightly above our alpha 0.05, showing that Area has some predictive value but is not as statistically significant as Elevation or Adjacent. Area also shows to have a pearson correlation with our response, suggesting that its relationship with Endemics may be linear.
 - Scrutz (p-val: 0.2044) and Nearest (p-val: 0.9066) have little to no meaningful relationship with Endemics that can be leveraged in prediction. Scrutz has a negative pearson correlation value of -0.15, which can suggest that islands further from Santa Cruz may have slightly less Endemics, but the value is weak and not statistically meaningful. Nearest has almost 0 pearson correlation with Endemics, and its high p-value means we fail to reject the null hypothesis that Nearest has no effect.

7. Predict the response value when all the chosen predictors take their median value. Quantify the prediction uncertainty.

Code:

```
gam_model_no_sqrt <- gam(Endemics ~ s(Area) + s(Elevation) + s(Scrutz) + s(Adjacent), data = gala)
median_values <- data.frame(
```

```

Area = median(gala$Area, na.rm = TRUE),
Elevation = median(gala$Elevation, na.rm = TRUE),
Scruz = median(gala$Scruz, na.rm = TRUE),
Adjacent = median(gala$Adjacent, na.rm = TRUE)
)

prediction <- predict(
  gam_model_no_sqrt,
  newdata = median_values,
  se.fit = TRUE
)

fit <- prediction$fit
se <- prediction$se.fit

alpha <- 0.05
z_value <- qnorm(1 - alpha / 2)
lower_ci <- fit - z_value * se
upper_ci <- fit + z_value * se

pred_uncertainty <- (upper_ci - lower_ci) / 2

cat("Predicted Endemics:", round(fit, 2), "\n")

```

```
## Predicted Endemics: 17.47
```

```
cat("95% Prediction Interval: [", round(lower_ci, 2), ",", round(upper_ci, 2), "]\n")
```

```
## 95% Prediction Interval: [ 10.53 , 24.4 ]
```

```
cat("Prediction Uncertainty: ", pred_uncertainty)
```

```
## Prediction Uncertainty: 6.936156
```

Answer:

- Prediction for Endemics for basic GAM model is 17.47, with a 95% Prediction Interval of [10.53, 24.4], and a Prediction uncertainty of 6.936.

Problem 2: Salary

```

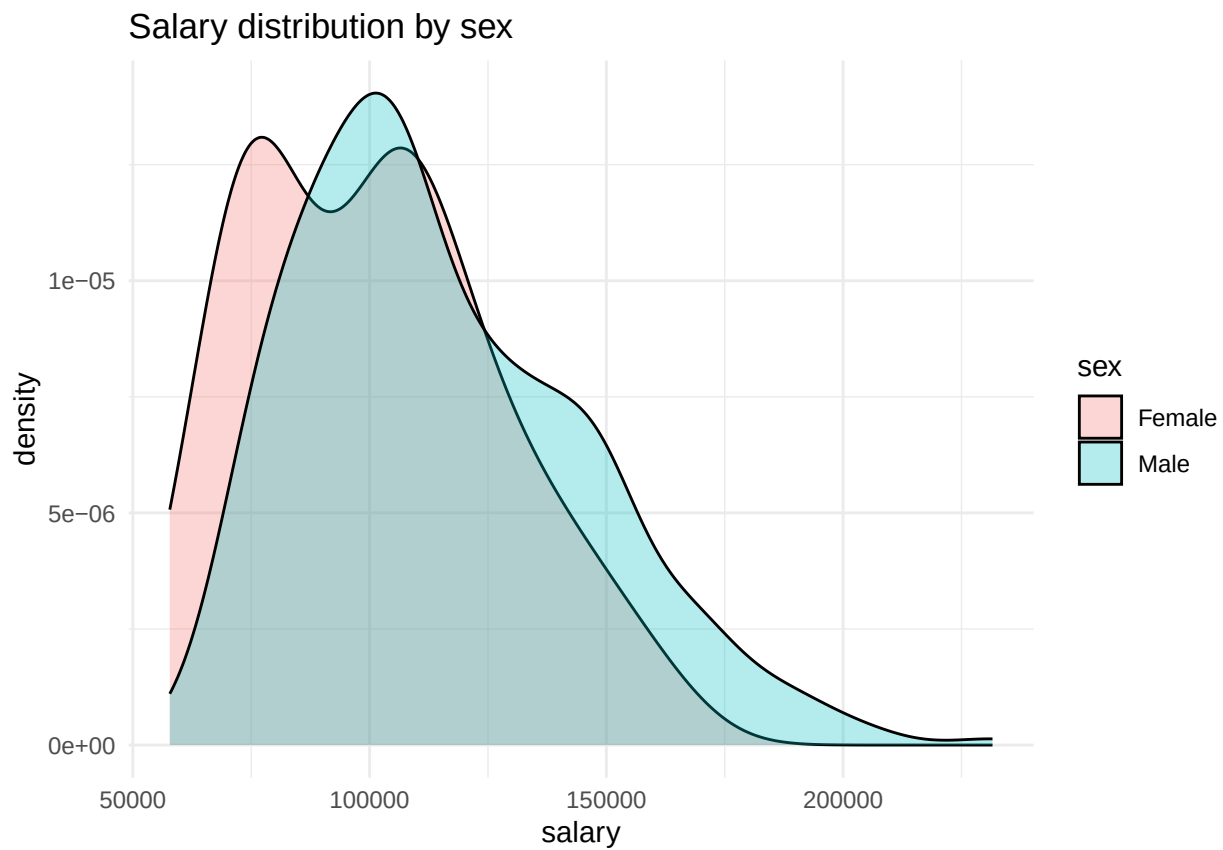
library(carData)
data(Salaries)

```

2. Comment on whether there is a need to include interactions between these categorical predictors. Select your model and justify your choice.

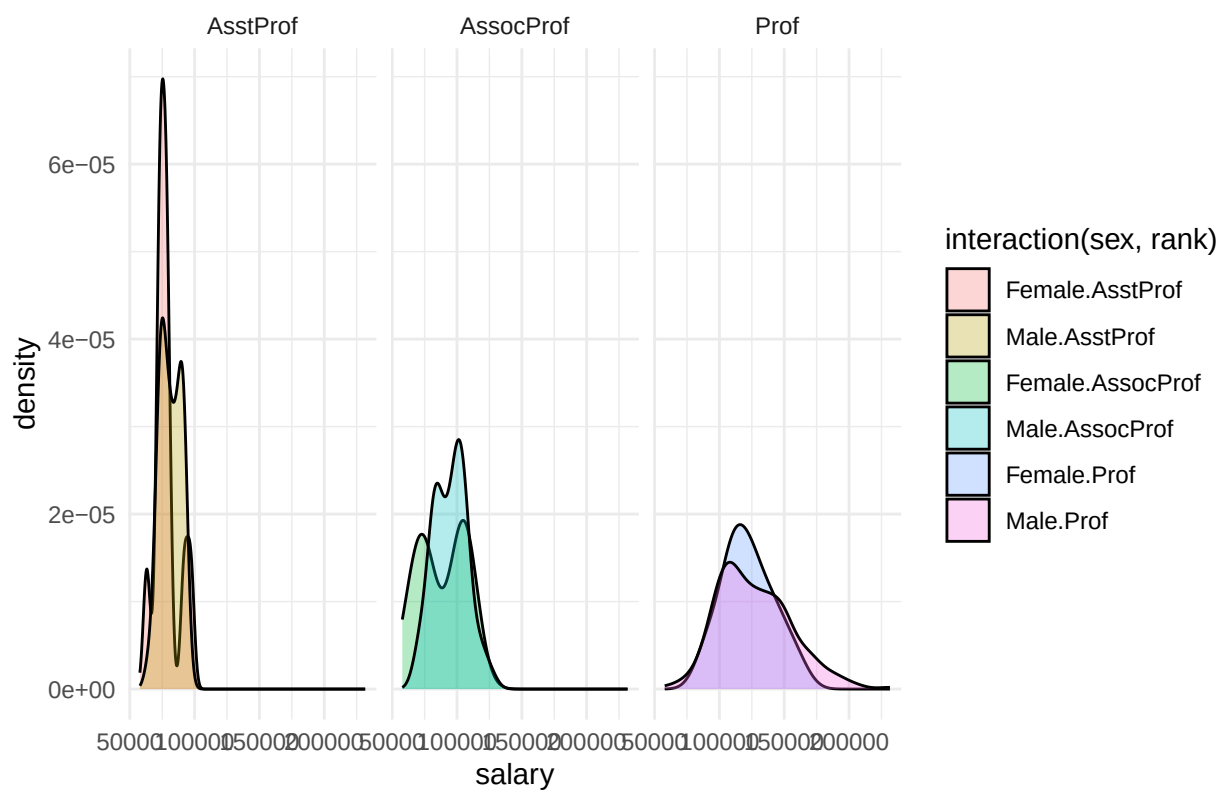
Code:

```
# salary dist by sex
ggplot(Salaries, aes(x = salary, fill = sex)) +
  geom_density(alpha = 0.3) +
  labs(title = "Salary distribution by sex", x = "salary", y = "density") +
  theme_minimal()
```



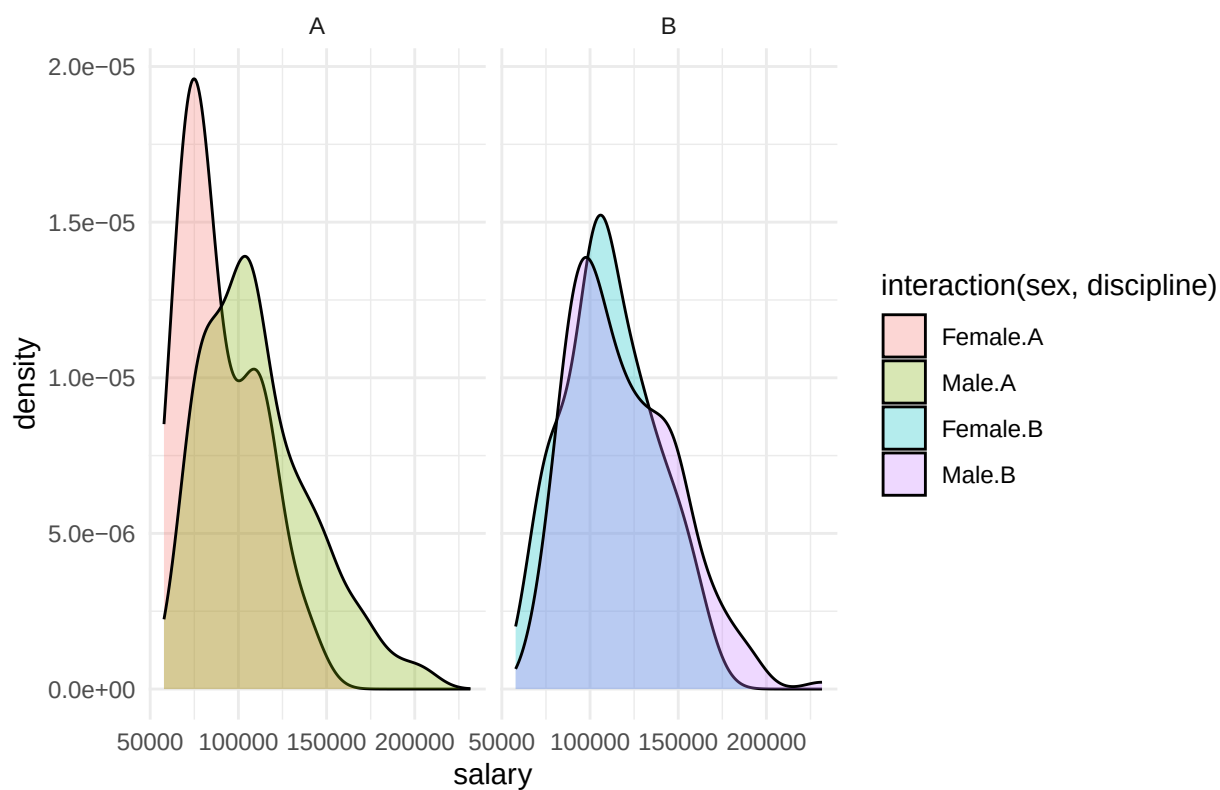
```
# sex and rank
ggplot(Salaries, aes(x = salary, fill = interaction(sex, rank))) +
  geom_density(alpha = 0.3) +
  facet_wrap(~rank) +
  labs(title = "Salary dist. Sex and Rank", x = "salary", y = "density") +
  theme_minimal()
```

Salary dist. Sex and Rank



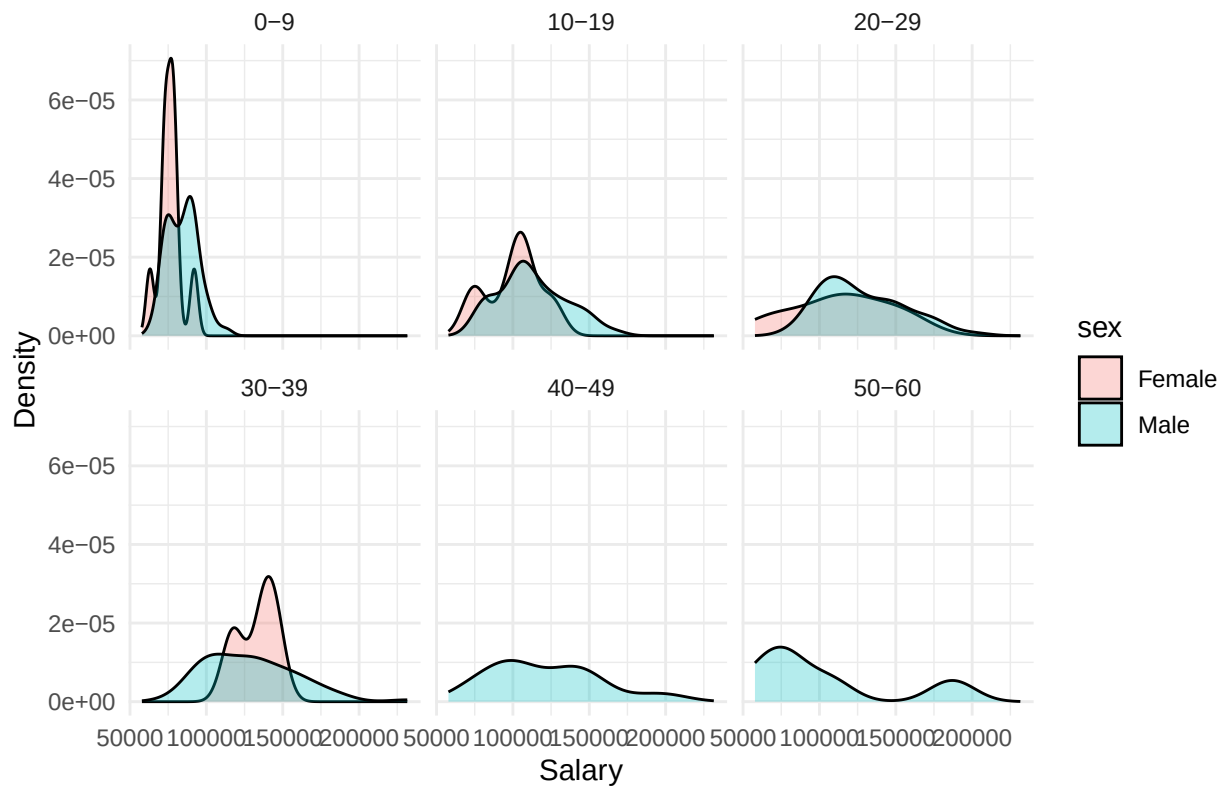
```
# sex and discipline
ggplot(Salaries, aes(x = salary, fill = interaction(sex, discipline))) +
  geom_density(alpha = 0.3) +
  facet_wrap(~discipline) +
  labs(title = "Salary dist. Sex and Discipline", x = "salary", y = "density") +
  theme_minimal()
```

Salary dist. Sex and Discipline



```
# salary dist: sex and yrs.since.phd
ggplot(Salaries, aes(x = salary, fill = sex)) +
  geom_density(alpha = 0.3) +
  facet_wrap(~ cut(yrs.since.phd, breaks = seq(0, 60, by = 10), right = FALSE,
    labels = c("0-9", "10-19", "20-29", "30-39", "40-49", "50-60"))) +
  labs(title = "Salary dist. Sex and yrs.since.phd (bin=10)", x = "Salary", y = "Density") +
  theme_minimal()
```


Salary dist. Sex and yrs.since.phd (bin=10)



```
# salary dist: sex, yrs.service
ggplot(Salaries, aes(x = salary, fill = sex)) +
  geom_density(alpha = 0.3) +
  facet_wrap(~ cut(yrs.service,
                   breaks = seq(0, 60, by = 10),
                   right = FALSE,
                   labels = c("0-9", "10-19", "20-29", "30-39", "40-49", "50-60")), drop = FALSE) +
  labs(title = "Salary dist. Sex and yrs.service (bin=10)", x = "Salary", y = "Count") +
  theme_minimal()
```

```
## Warning: Groups with fewer than two data points have been dropped.
## Groups with fewer than two data points have been dropped.
```

```
## Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning
## -Inf
## Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning
## -Inf
```

Salary dist. Sex and yrs.service (bin=10)



```
# contingency table for pairwise
sex_rank <- table(Salaries$sex, Salaries$rank)
sex_disc <- table(Salaries$sex, Salaries$discipline)
rank_disc <- table(Salaries$rank, Salaries$discipline)
```

```
sex_rank
```

```
##
##           AsstProf AssocProf Prof
##   Female         11         10   18
##   Male          56         54  248
```

```
sex_disc
```

```
##
##           A    B
##   Female  18  21
##   Male   163 195
```

```
rank_disc
```

```
##
##           A    B
##   AsstProf  24  43
##   AssocProf 26  38
##   Prof     131 135
```

```
# chi test for association
chisq.test(sex_rank)
```

```
##
## Pearson's Chi-squared test
##
## data: sex_rank
## X-squared = 8.5259, df = 2, p-value = 0.01408
```

```
chisq.test(sex_disc)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: sex_disc
## X-squared = 2.0875e-29, df = 1, p-value = 1
```

```
chisq.test(rank_disc)
```

```
##
## Pearson's Chi-squared test
##
## data: rank_disc
## X-squared = 4.6487, df = 2, p-value = 0.09785
```

- (Sex, Rank):
 - In the event that salary disparities between Male and Female are more pronounced at higher ranks (like full professors compared to assistant professors) it becomes necessary to explore an interaction between 'sex' and 'rank'. Additionally, our Pearson Chi-squared test returns a p-value of $0.01408 < \alpha = 0.05$, confirming significant association between sex and rank.
- (Sex, Discipline):
 - Our Pearson Chi-squared for this pairwise returns a p-value of 1, significantly greater than our alpha. However, in the event that 'Male' and 'Female' staff face salary discrepancies among fields of study, with more significant gaps apparent in 'applied' disciplines versus 'theoretical', it becomes essential to consider an interaction between discipline and sex.
- To account for all confounding variables like 'rank', 'discipline', and 'yrs.since.phd'/'yrs.service', I will use a basic linear regression model:

$\text{Salary} \sim \text{sex} + \text{rank} + \text{discipline} + \text{yrs.since.phd} + \text{yrs.service}$

- I will also establish a secondary model to test for overall salary differences for sex with focus on interaction between confounding variables 'rank' and 'discipline'.

$\text{Salary} \sim \text{sex} * \text{rank} + \text{sex} * \text{discipline} + \text{yrs.since.phd} + \text{yrs.service}$

```

model_basic <- lm(salary ~ sex + rank + discipline + yrs.since.phd + yrs.service, data=Salaries)
model_rank_disc <- lm(salary ~ sex * rank + sex * discipline + yrs.since.phd + yrs.service, data=Salaries)

summary(model_basic)

```

```

##
## Call:
## lm(formula = salary ~ sex + rank + discipline + yrs.since.phd +
##     yrs.service, data = Salaries)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -65248 -13211  -1775   10384   99592
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   65955.2     4588.6   14.374 < 2e-16 ***
## sexMale       4783.5     3858.7    1.240  0.21584
## rankAssocProf 12907.6     4145.3    3.114  0.00198 **
## rankProf      45066.0     4237.5   10.635 < 2e-16 ***
## disciplineB   14417.6     2342.9    6.154 1.88e-09 ***
## yrs.since.phd  535.1       241.0    2.220  0.02698 *
## yrs.service   -489.5       211.9   -2.310  0.02143 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22540 on 390 degrees of freedom
## Multiple R-squared:  0.4547, Adjusted R-squared:  0.4463
## F-statistic: 54.2 on 6 and 390 DF, p-value: < 2.2e-16

```

```

summary(model_rank_disc)

```

```

##
## Call:
## lm(formula = salary ~ sex * rank + sex * discipline + yrs.since.phd +
##     yrs.service, data = Salaries)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -64874 -13468  -1382   9888   99935
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   66450.1     7670.5   8.663 < 2e-16 ***
## sexMale       4002.5     8317.9    0.481  0.63065
## rankAssocProf  6466.1    10010.8    0.646  0.51872
## rankProf     39243.2     9006.7   4.357 1.69e-05 ***
## disciplineB   21616.0     7323.3   2.952  0.00335 **
## yrs.since.phd  540.9       241.6    2.239  0.02572 *
## yrs.service   -500.5       212.6   -2.355  0.01903 *
## sexMale:rankAssocProf  7479.0    10820.1    0.691  0.48985
## sexMale:rankProf    6639.4     9326.3    0.712  0.47695

```

```
## sexMale:disciplineB    -7873.2      7693.9  -1.023  0.30680
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22580 on 387 degrees of freedom
## Multiple R-squared:  0.4569, Adjusted R-squared:  0.4442
## F-statistic: 36.17 on 9 and 387 DF,  p-value: < 2.2e-16
```

```
# R-squared to test accuracy
print(c("Basic Model adjusted R-squared: ", summary(model_basic)$adj.r.squared))
```

```
## [1] "Basic Model adjusted R-squared: " "0.446287031864428"
```

```
print(c("Rank/Discipline interaction Model adjusted R-squared: ", summary(model_rank_disc)$adj.r.squared))
```

```
## [1] "Rank/Discipline interaction Model adjusted R-squared: "
## [2] "0.444226559508176"
```

```
# AIC to test fit/complexity balance
AIC(model_basic, model_rank_disc)
```

```
##           df      AIC
## model_basic      8 9093.826
## model_rank_disc 11 9098.235
```

```
BIC(model_basic, model_rank_disc)
```

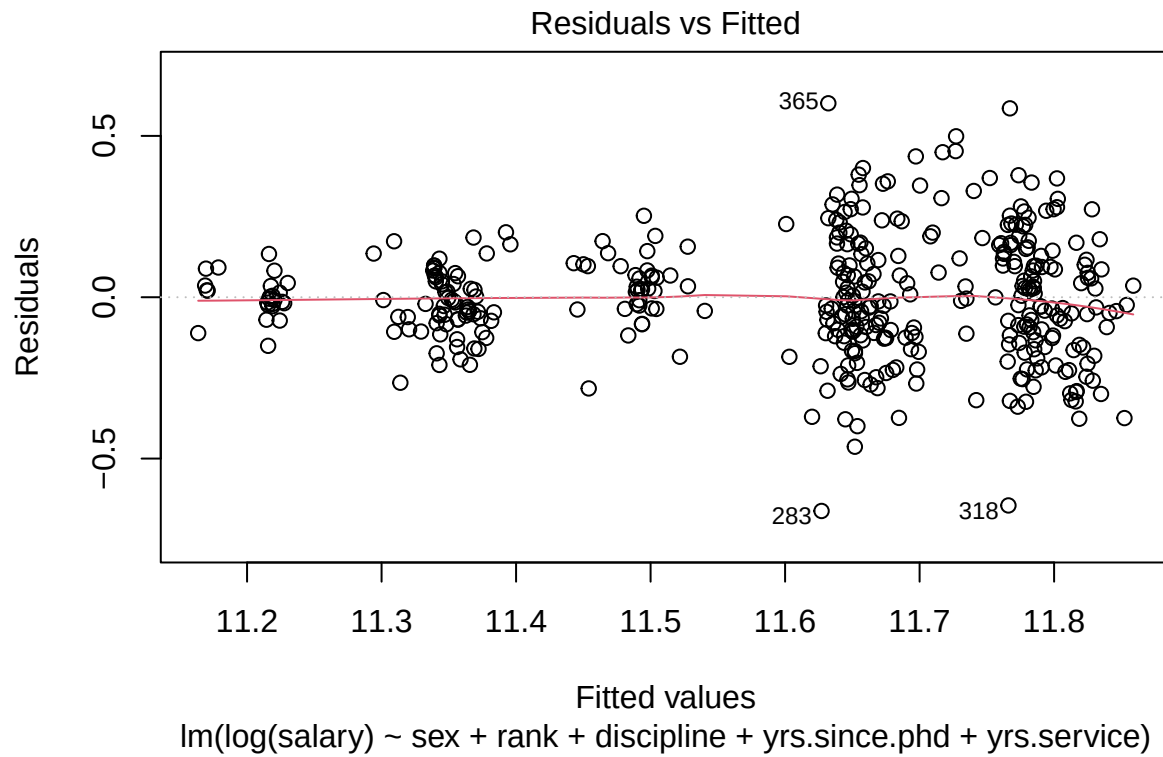
```
##           df      BIC
## model_basic      8 9125.698
## model_rank_disc 11 9142.058
```

- As our basic model shows the best values for adjusted R-squared, and also shows improvement in AIC/BIC criterion values with lower complexity (df=8), it is the better of the two models.

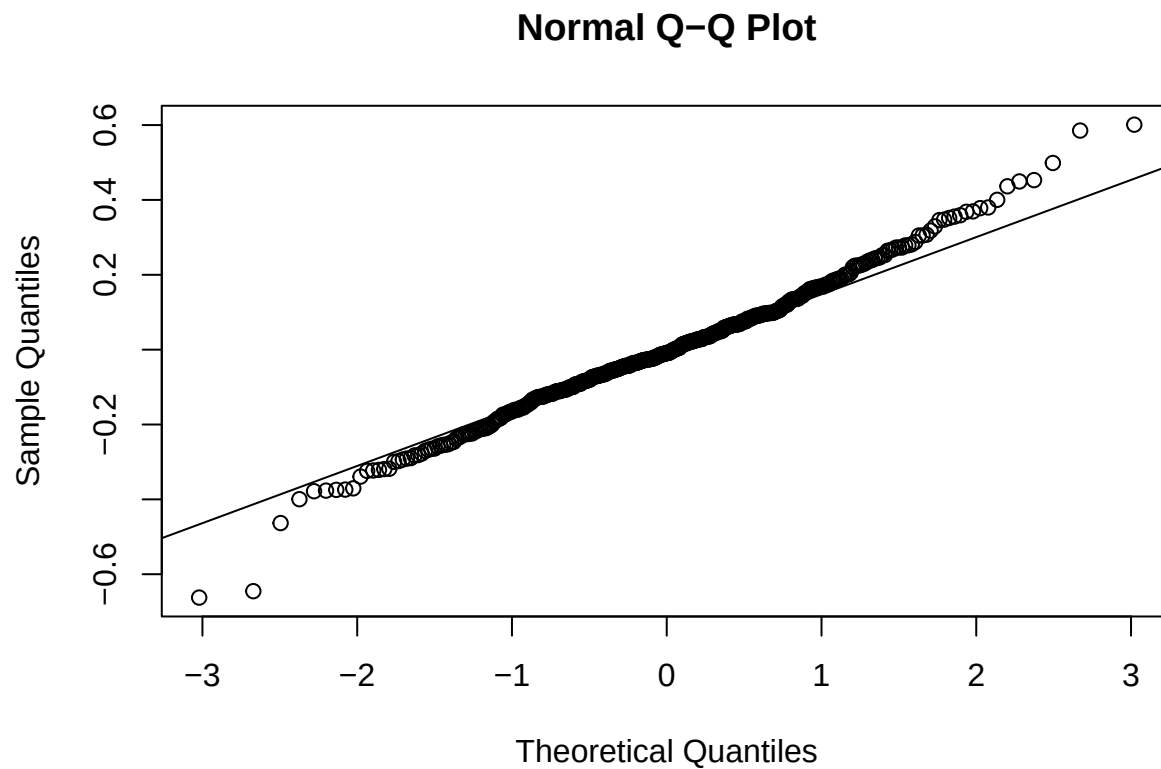
4. How well does the chosen model fit the data? Quantify the fit numerically. Additionally, check the model assumptions and summarize your findings.

- Our model fit improved with the log transformation to response variable 'salary', with significantly improved Akaike and Bayesian Information Criterion values of [A:-222.78, B:-190.91] as opposed to the prior [A:9093.83, B:9125.7].

```
model_basic_log <- lm(log(salary) ~ sex + rank + discipline + yrs.since.phd + yrs.service, data=Salaries)
# Checking model assumptions
# residual vs fitted
plot(model_basic_log, which = 1)
```



```
# qq plot
qqnorm(resid(model_basic_log))
qqline(residuals(model_basic_log))
```



- Residuals vs. Fitted

- Residuals appear randomly scattered around 0.0, meets assumption of linearity.
 - Visible clustering reveals variance may not be constant, mildly violating heteroscedasticity assumption.
- QQ Plot
 - Shows residuals to follow a normal distribution, as most points lie near our reference line.
 - Slight deviations on tail indicates mild non-normality of residuals, likely due to outliers.
- Our model meets assumptions of linearity and normality, however there may be some non-constant variance as well as a small amount of outliers.